



MGT 207

Inventory Management

What is Inventory?

Inventory: A stock of materials used to satisfy customer demand or support production.



Key Purposes

- **Customer Demand:** Ready to fulfill orders immediately.
- **Production Support:** Supplies for manufacturing goods or providing services.
- **Operational Buffer:** Managing variability in supply chains.

Pressures for Small Inventories



Inventory Holding Cost

The primary pressure to keep inventory low is the significant expense of maintaining stock over time.

Cost Components

-  Cost of Capital
-  Storage & Handling
-  Taxes & Insurance

Inventory Shrinkage

- Pilferage (Theft)
- Obsolescence
- Deterioration

Pressures for Large Inventories



Drivers for Higher Stock

There are several key operational and financial pressures that encourage businesses to maintain larger inventory levels.

Key Pressures

- + Customer Service & Stockouts
- Ordering & Setup Costs
- Labor & Equipment Efficiency
- Transportation Economies

Payments to Suppliers

- Quantity discounts on purchase price

Types of Inventory

Operational Inventories



Cycle Inventory

Inventory that results from the lot size or batch size used in production or purchasing.



Safety Stock

Buffer stock held to protect against uncertainties in demand, lead time, or supply.



Pipeline Inventory

Inventory that is in transit between different locations in the supply chain.

Cycle Inventory

Lot Sizing Principles



The lot size, Q , varies directly with the elapsed time (or cycle) between orders.



The longer the time between orders for a given item, the greater the cycle inventory must be.

Cycle Inventory Calculation

$$\text{Average cycle inventory} = \frac{Q + 0}{2} = \frac{Q}{2}$$

Pipeline Inventory

Components and Definitions



Lead Time Demand

(Average) demand during lead time.

 D_L 

Periodic Demand

(Average) demand for the items per period.

 $\bar{d}$ 

Lead Time Duration

Number of periods in the item's lead time.

 L

Pipeline Inventory Calculation

$$\text{Pipeline inventory} = D_L = \bar{d} L$$

Example 1 (1 of 3)

Scenario Details

A plant makes monthly shipments of electric drills to a wholesaler.

The wholesaler pays for inventory from the moment the plant makes a shipment.

The Challenge:

What is the effect on the wholesaler's cycle and pipeline inventories?

Current vs. Proposed

CURRENT STATUS

- Lot Size: **280 drills**
- Demand: **70 drills/week**
- Lead Time: **3 weeks**

NEW PROPOSAL

- Lot Size: **350 units**
- Demand: **70 drills/week**
- Lead Time: **2 weeks (Guaranteed)**

Example 1 (2 of 3)

The wholesaler's current cycle and pipeline inventories are calculated as follows:

Cycle Inventory

Formula: $Q / 2$

Calculation:

140 drills

(280 / 2)

Pipeline Inventory

Formula: $D_L = \bar{d}L$

Calculation:

(70 drills/week) (3 weeks)

210 drills

Example 1 (3 of 3)

The wholesaler's cycle and pipeline inventories if they accept the new proposal:

Cycle Inventory

Formula: $Q / 2$

Calculation:

175 drills

(350 / 2)

Pipeline Inventory

Formula: $D_L = \bar{d}L$

Calculation:

(70 drills/week) (2 weeks)

140 drills

Safety Stock Inventory

Extra inventory held to protect against demand or supply uncertainty.

(We will calculate it later in the probabilistic EOQ model.)



- ▲ Now that we understand the three main types, let's look at the costs that drive decisions about them.

Inventory Costs: The Two Big Ones

Holding (Carrying) Cost (H)

- Cost of keeping one unit in inventory for one year (capital, storage, obsolescence, etc.).

Ordering / Setup Cost (S)

- Fixed cost every time we place an order or set up a production run (independent of order size).



- ▲ These two costs are in direct conflict — this fundamental tension creates the need for the EOQ model.

Economic Order Quantity (1 of 2)

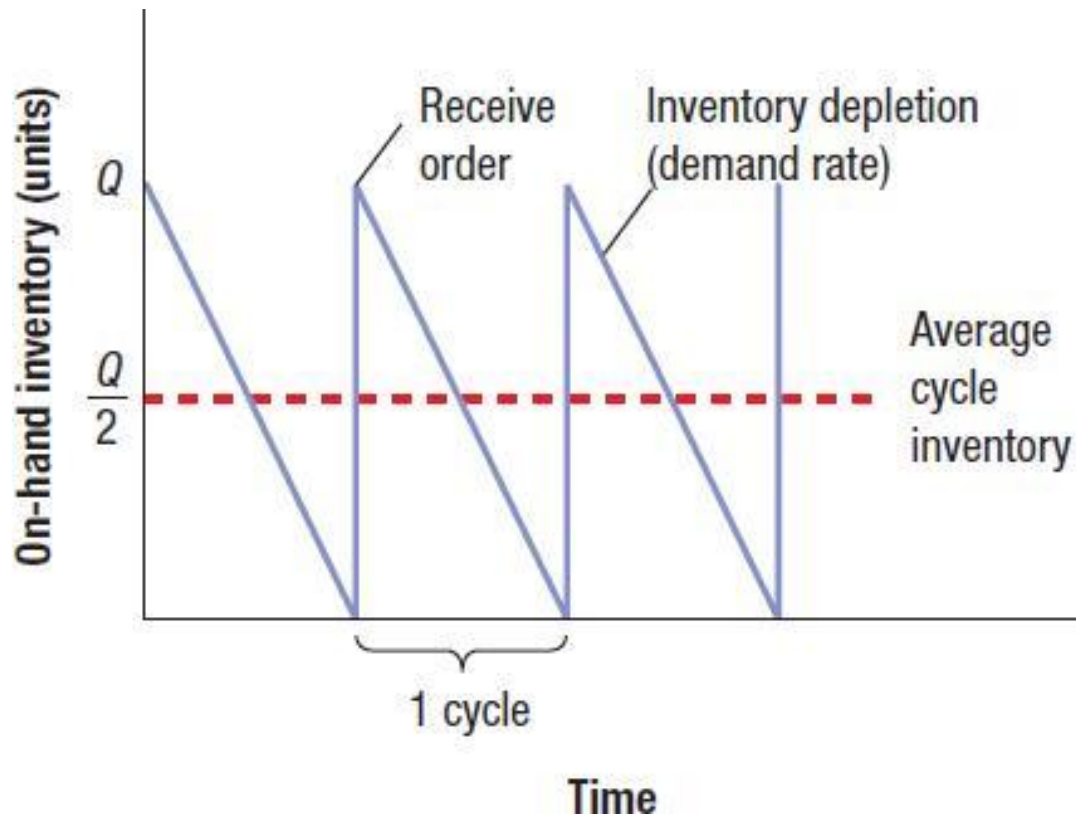
The lot size, Q , that minimizes total annual inventory holding and ordering costs.

Five Key Assumptions

- 1 The demand rate is constant and known with certainty (will be relaxed).
- 2 No constraints are placed on the size of each lot.
- 3 The only two relevant costs are the inventory holding cost and the fixed cost per lot for ordering or setup.
- 4 Decisions for one item can be made independently of decisions for other items.
- 5 The lead time is constant and known with certainty (will be relaxed).

Calculating EOQ (1 of 5)

Figure 9.5 Cycle-Inventory Levels



Calculating EOQ (2 of 5)

- ▶ Annual holding cost

$$\text{Annual holding cost} = (\text{Average cycle inventory}) \times (\text{Unit holding cost})$$

- ▶ Annual ordering cost

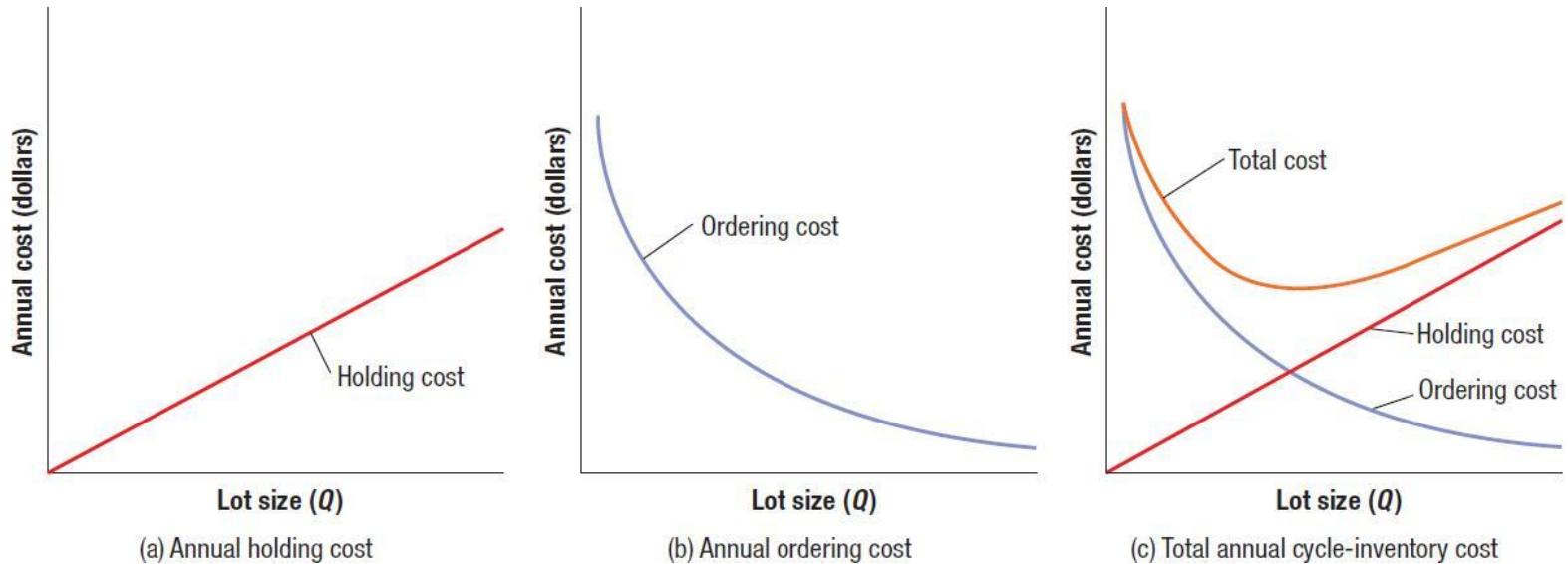
$$\text{Annual ordering cost} = (\text{Number of orders/Year}) \times (\text{Ordering or setup costs})$$

- ▶ Total cost

$$\text{Total cost} = \text{Annual holding cost} + \text{Annual ordering or setup cost}$$

Calculating Total Inv Cost

Figure 9.6 Graphs of Annual Holding, Ordering, and Total Costs



Total Inventory Cost

Total cost

$$C = \frac{Q}{2}(H) + \frac{D}{Q}(S)$$

where

C = total annual cycle-inventory cost

Q = lot size (in units)

H = holding cost per unit per year

D = annual demand (in units, per year)

S = ordering or setup costs (in dollars per lot)

Example 2 (1 of 3)

Problem Context



- Museum gift shop operates **52 weeks/year**.
- Top-selling SKU: **Bird feeder**.
- Current policy: **468-unit lot size**.

Cost Parameters



Sales (d): 18 units / week

Supplier Charge (P): \$60 / unit

Ordering Cost (S): \$45 / order

Holding Cost (H): 25% of feeder's value

Key Questions to Solve:

1. What is the annual cycle-inventory cost of the current 468-unit lot size policy?
2. Would a lot size of **390** be better?

Example 2 (2 of 3)

Step 1: Compute Annual Demand (D) and Holding Cost (H)

$$D = (18 \text{ units / week})(52 \text{ weeks / year}) = \mathbf{936 \text{ units}}$$

$$H = 0.25(\$60 / \text{unit}) = \mathbf{\$15}$$

Alternative Lot Size (Q = 390)

Total annual cycle-inventory cost:

$$\begin{aligned} C &= \frac{Q}{2}(H) + \frac{D}{Q}(S) = \frac{390}{2}(\$15) + \frac{936}{390}(\$45) \\ &= \$2,925 + \$108 = \mathbf{\$3,033} \end{aligned}$$

Current Policy (Q = 468)

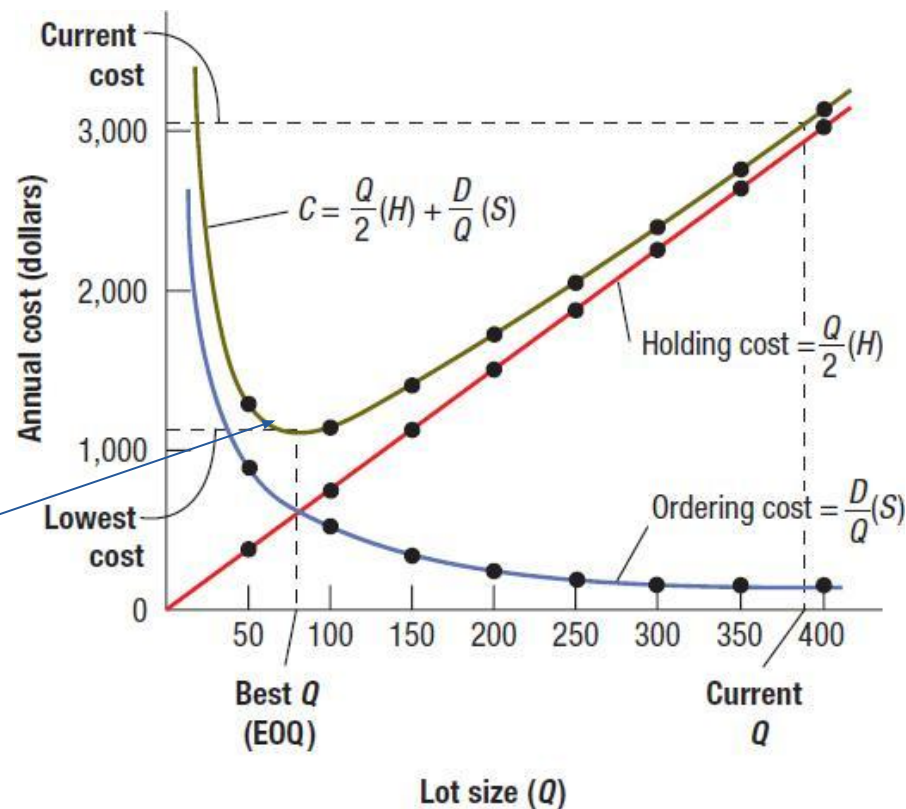
Total annual cycle-inventory cost:

$$C = \frac{468}{2}(\$15) + \frac{936}{468}(\$45) = \$3,510 + \$90 = \mathbf{\$3,600}$$

The alternative lot size (Q = 390) results in a lower annual cost of \$3,033 compared to \$3,600.

Example 2 (3 of 3)

Total Annual Cycle-Inventory Cost Function for the Bird Feeder



Min total cost reached when the two costs cross

Deriving the EOQ Formula

Objective: Minimize Total Annual Cost (TC)

The optimal order quantity Q^* occurs where the sum of annual holding and ordering costs is at its minimum.

The Derivation Steps:

- Set Annual Holding Cost equal to Annual Ordering Cost.
 - $(Q / 2) * H = (D / Q) * S$
- Multiply both sides by Q to consolidate the variable:
 - $(Q^2 / 2) * H = D * S$
- Isolate Q^2 by multiplying by 2 and dividing by H :
 - $Q^2 = (2 * D * S) / H$
- Take the square root of both sides to find the optimal Q :

Economic Order Quantity

$$Q = \sqrt{\frac{2DS}{H}}$$

Q = Optimal order quantity
 D = Annual demand in units
 S = Order cost per purchase order
 H = Annual holding cost per unit

Note: This derivation assumes the total cost curve is convex, where the first derivative equals zero.

EOQ Review So Far...

1. Economic Order Quantity (EOQ)

$$EOQ = \sqrt{\frac{2DS}{H}}$$

2. Time Between Orders (TBO)

$$TBO_{EOQ} = \frac{EOQ}{D} (12 \text{ months / year})$$

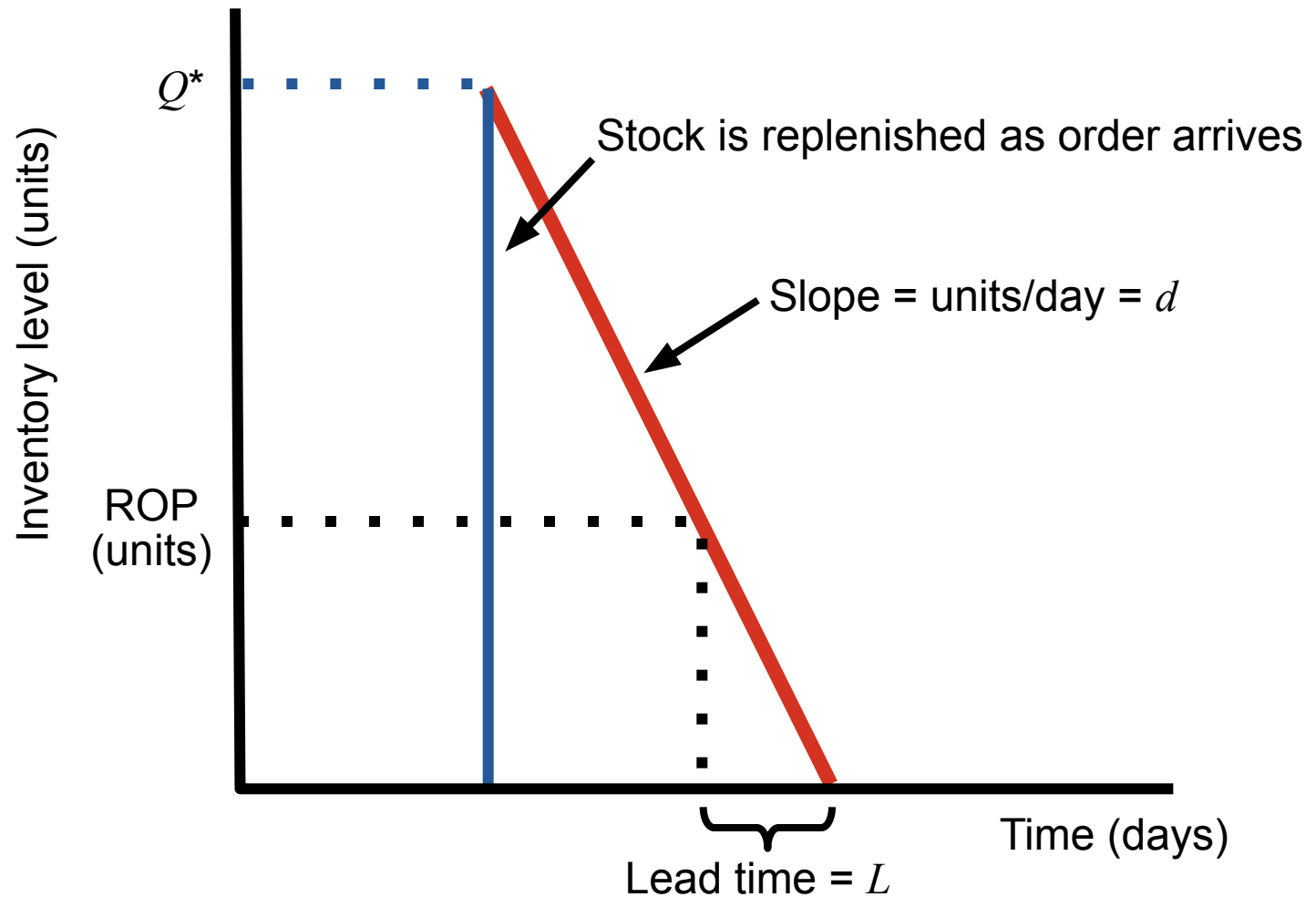
The Missing Piece:

**When do you
order?**

What is missing to implement
inventory management using EOQ?



Reorder Point Curve



Reorder Points

- ▶ The reorder point (ROP) tells “when” to order
- ▶ Lead time (L) is the time between placing and receiving an order

$$\text{ROP} = \left(\begin{array}{c} \text{Demand} \\ \text{per day} \end{array} \right) \left(\begin{array}{c} \text{Lead time for a new} \\ \text{order in days} \end{array} \right)$$

$$\text{ROP} = d \times L$$

$$d = \frac{D}{\text{Number of working days in a year}}$$

Reorder Point Example

Demand = 8,000 units per year

250 working day year

Lead time for orders is 3 working days, may take 4

$$d = \frac{D}{\text{Number of working days in a year}}$$

$$= 8,000/250 = 32 \text{ units}$$

$$\text{ROP} = d \times L$$

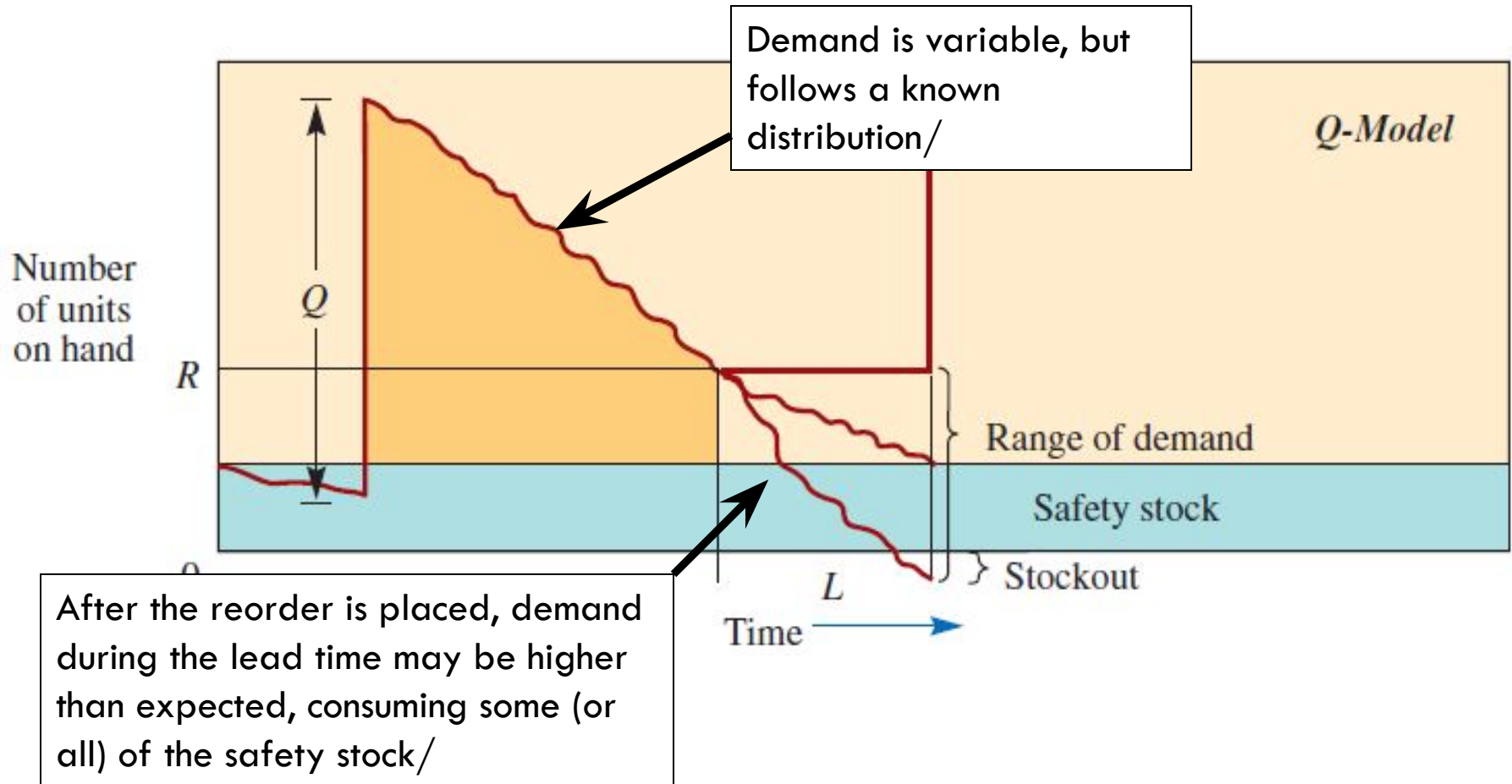
$$= 32 \text{ units per day} \times 3 \text{ days} = 96 \text{ units}$$

$$= 32 \text{ units per day} \times 4 \text{ days} = 128 \text{ units}$$

Probabilistic Fixed-Order Quantity (EOQ) Model

- ▶ In the fixed order quantity model, the ordering process is triggered when the inventory level drops to a critical point, the *Reorder Point (R)*.
- ▶ This starts the *lead time* for the item.
- ▶ During the lead time, customers continue to draw down the inventory
- ▶ If demand is uncertain, during this period the inventory is vulnerable to stockout (run out of inventory)

Inventory on hand



Probabilistic Fixed-Order Quantity (EOQ) Model

- ▶ The customer **service level** is converted into a **z** value using the standard normal distribution table
- ▶ *Safety Stock:*

$$SS = z \sigma_L = z \sqrt{L} \sigma_d$$

- ▶ *Expected Demand During Lead Time:*

$$EDDLT = \bar{d}L$$

- ▶ *Reorder Point:*

$$R = EDDL T + SS = \bar{d}L + z \sqrt{L} (\sigma_d)^2$$

Probabilistic Fixed-Order Quantity (EOQ) Model

Toy Inventory at **Sell-Rite Discount Stores**. Data indicates a **constant lead time (L) of 10 days** and a **normally distributed demand** with a mean of **1,250 toys/day** and a standard deviation of **375 toys/day**.

a. What is the *EDDLT* for the toy?

Expected Demand During Lead Time calculation.

b. What is the standard deviation of *DDLT* for the toy?

Standard deviation of Demand During Lead Time.

c. Compute the reorder point (ROP) for the toy.

Service level specified at 90 percent during lead time.

d. What is the safety stock?

The buffer inventory held to prevent stockouts.

Probabilistic Fixed-Order Quantity (EOQ) Model

$L = 10, \bar{d} = 1,250 \text{ toys}, \sigma_d = 375 \text{ toys}$

a. $EDDLT = \bar{d}L = 10(1,250) = 12,500 \text{ toys}$

b. $\sigma_L = \sqrt{L\sigma_d^2} = 375\sqrt{10} = 1,186 \text{ toys}$

c. From the Normal table, using a service level of 90% or 0.9, we obtain $z = 1.28$.

Therefore,

$$\begin{aligned} R &= EDDL T + z\sigma_L = 12,500 + 1.28(1,186) \\ &= 14,018 \text{ toys.} \end{aligned}$$

d. $SS = z\sigma_L = 1.28(1,186) = 1,518 \text{ toys.}$

Transitioning: From EOQ to Newsvendor

Moving from deterministic cost minimization to betting on uncertain demand

EOQ Model

Cost-Based Logic

Focuses on finding the optimal balance between **holding costs** and **ordering costs**.

- Constant, known demand
- Deterministic lead times
- Goal: Minimize total annual cost

Newsvendor

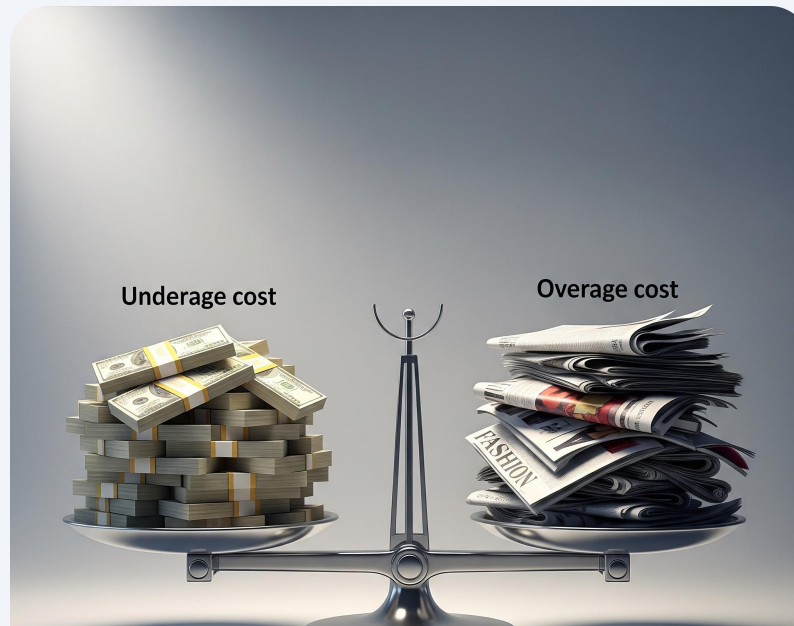
Inventory Betting

A "one-shot" decision under **uncertain demand** where the risk of overstocking meets the cost of lost sales.

- Stochastic, random demand
- Perishable/Short-lifecycle goods
- Goal: Maximize expected profit

The Trade-off

Overage vs. Underage Cost



The Risk

Waste vs. Lost Opportunities

Inventory Model for Perishable Goods

Betting on Demand

Betting on Demand

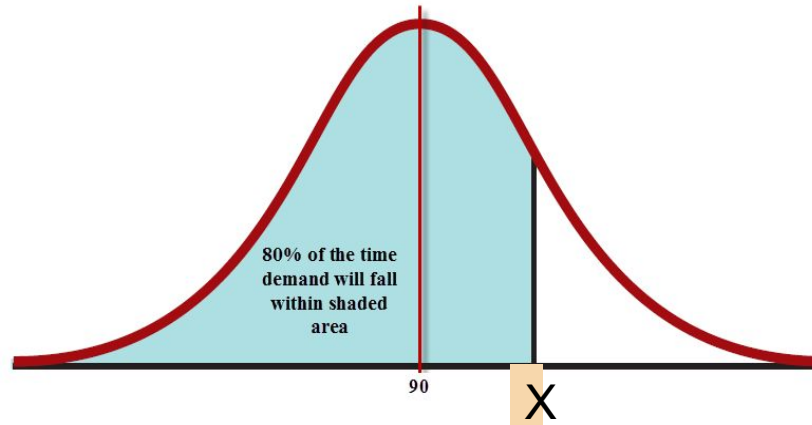
Consider the problem of deciding how many loaves of bread to bake every day

Too few and some customers will not be able to purchase bread, and profits associated with these potential sales are lost.

Too many loaves lead to waste

How many loaves should we stock to ensure 80% service level?

- Suppose that demand follows a Normal distribution with a mean of 90 loaves and a standard deviation of 10 loaves



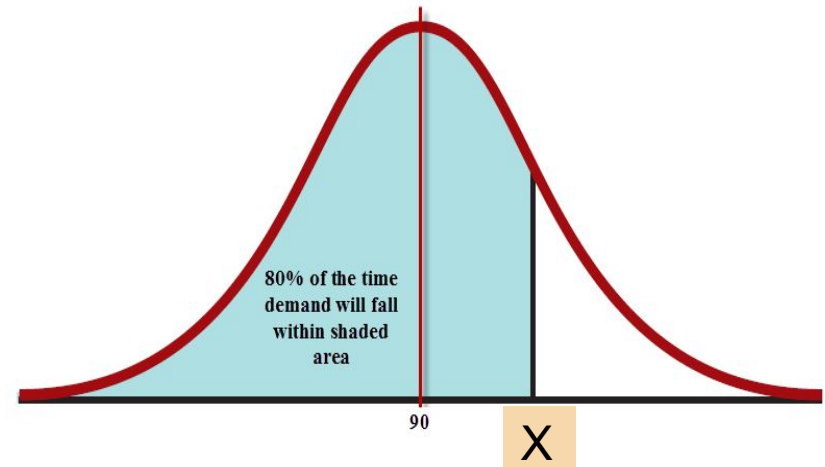
- Stocking 90 loaves (average demand) yields a 50% service level (chance of not stocking out)
- We want an 80% service level (chance of not stocking out), i.e. $P=80\%$.

Solving the Stocking Problem

G(z)	z
0.50	0
0.55	0.126
0.60	0.253
0.65	0.385
0.70	0.524
0.75	0.674
0.80	0.842
0.85	1.036

- From z-table, 0.842 'standard deviation' of 'extra loaves' are needed (i.e., 0.842 standard deviation away from mean demand 90).
- Standard deviation is 10.
- Number of *extra loaves* = $0.842 \times 10 \cong 8$ loaves

Excel: "`=NORMSINV(0.8)`" $\cong 0.842$



Pricing Information

- ▶ Suppose that:
 - ▶ Each loaf sells for \$4.50
 - ▶ The baker incurs \$1.50 in unit cost
 - ▶ Day-old bread can be sold for \$0.50 per loaf
- ▶ Is the optimal stocking level still $90+8=98$?

Optimal service level P

There is an optimal way of choosing
service level P

$$P = \frac{C_u}{C_o + C_u}$$

$C_o = (C - S)$: unit overage cost

$C_u = (P - C)$: unit underage cost

In Our Case

▣ $C_u = \$4.50 - \$1.50 = \$3.00$

▣ $C_o = \$1.50 - \$0.50 = \$1.00$

▣ $C_u + C_o = \$4.00$

$$C_u = \$3 \quad C_o = \$1$$

$$P = \frac{C_u}{C_o + C_u} = \frac{3}{1 + 3} = 75\%$$

- So, 80% was not optimal, 75% is. We would incur a lower expected cost with 75% service level than with 80% service level.

Solving the Stocking Problem

G(z)	z
0.50	0
0.55	0.126
0.60	0.253
0.65	0.385
0.70	0.524
0.75	0.674
0.80	0.842
0.85	1.036

- From z-table, 0.674 'standard deviation' of 'extra loaves' are needed (i.e., 0.674 standard deviation away from mean demand 90).
- Standard deviation is 10.
- Number of *extra papers* = $0.674 \times 10 \cong 6$ loaves
- So, the stocking quantity is $90 + 6 = 96$ loaves

Optimizing the Newsvendor Inventory Policy

Balance overage and underage costs to achieve the target service level.



01. The Economic Trade-off

Underage Cost (Cu): Profit lost per unit of unmet demand.

Overage Cost (Co): Loss incurred for each unsold unit at the end of the period.

02. Critical Ratio (P)

The optimal service level is the probability of not stocking out:

$$P = C_u / (C_o + C_u)$$

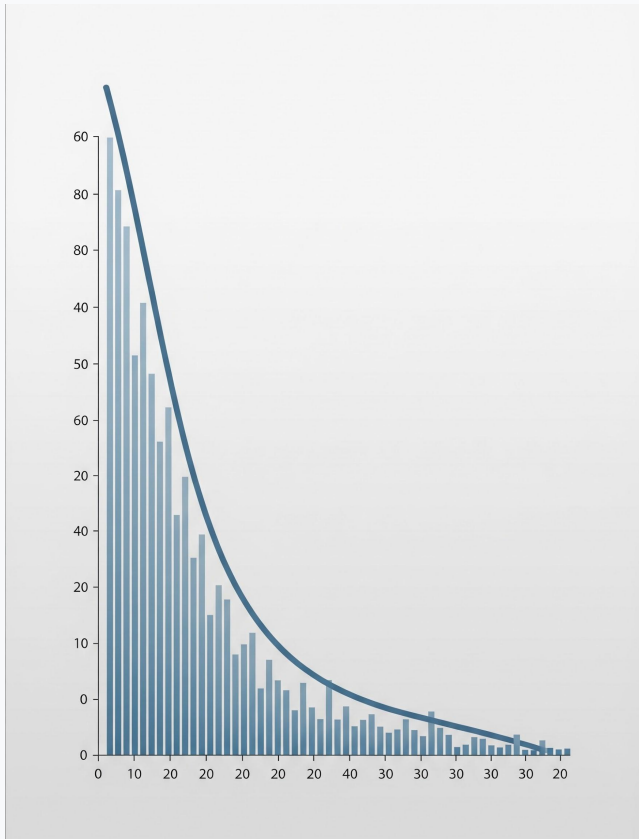
03. Stocking Decision

Convert the target service level **P** into a **z-value** using the standard normal distribution table to determine the optimal quantity **Q***.

Sources: Optimal service level P, Pricing Information

Handling Non-Normal Demand Distributions

The Critical Ratio method remains valid even when demand isn't Gaussian.



01. Discrete Distributions

If demand is discrete (e.g., individual items), find the **smallest quantity Q** such that the cumulative probability:

$$F(Q) \geq P \Leftrightarrow \text{Prob}\{D \leq Q\} \geq \text{Service Level}$$

02. Empirical Distributions

Use **historical data** directly to create a cumulative frequency table.

Choose the quantity corresponding to the target percentile **P** .

Uncertainty in Demand and Lead Times

Transitioning from fixed assumptions to probabilistic modeling.



Deterministic Models

Assume **exact knowledge** of demand and lead time. Everything is a fixed constant.

Stochastic Models

Demand and lead time become **random variables**. We manage the probability of outcomes.

Primary Adaptation

Treat **Lead Time Demand (LTD)** as a random variable to determine safety stocks and reorder points.

Motivating Case: Car Dealership Inventory

Balancing luxury stock costs against the risk of lost high-value sales.



Scenario: Julia's Mercedes Dealership

- Monthly sales range from **6 to 12 cars**.
- Lead times vary between **1 to 4 months**.
- Current Policy: Order **14 cars** when stock hits **12**.
- Goal: Determine the optimal order quantity (Q) and reorder point (R) to minimize total costs.

The High Stakes of Uncertainty

Luxury inventory management involves significant cost trade-offs:

- **Carrying Cost: \$600** /car/month.
- **Ordering Cost: \$570** /order.

Stochastic modeling is required because both demand and delivery times are uncertain.

The (Q, r) Continuous-Review Policy

Optimizing replenishment for random demand and lead time scenarios.



Policy Fundamentals

- Monitor inventory position **continuously**.
- Order size **Q** when position hits reorder point **r**.
- Addresses **random demand** and **random lead time**.
- Balances ordering costs, holding costs, and stockout risk.

Key Performance Measures

Average Inventory:

$$\sim Q/2 + \text{Safety Stock}$$

Cycle Service Level:

The probability of no stockout occurring during a single replenishment cycle.

Order Quantity Q: The EOQ Foundation

Applying the Economic Order Quantity formula in stochastic environments.



The Classic EOQ Formula

$$Q = \sqrt{(2DS / H)}$$

D: Avg Annual Demand

S: Fixed Setup Cost

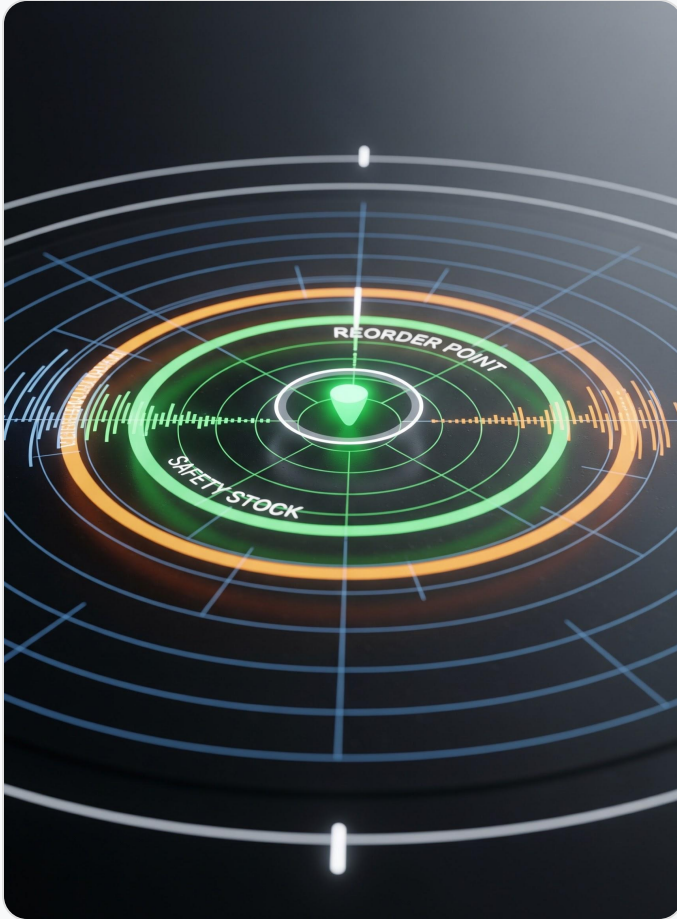
H: Holding Cost per Unit/Year

Why It Still Works

- **Cost Optimization:** It mathematically minimizes the sum of ordering costs and holding costs.
- **Stochastic Validity:** The formula remains valid even with random demand, treating it as a renewal process.
- **Decoupled Variables:** Lead-time variability does not affect Q; it is handled separately by the reorder point (r).
- **Robustness:** EOQ is notoriously "flat" around the optimum, meaning small errors in estimation don't significantly raise total costs.

Reorder Point r: Managing Uncertainty

Defining the inventory level that triggers a replenishment order.



The Reorder Point Formula

$$r = \mu_{DLT} + \text{Safety Stock}$$

$$\mu_{DLT} = \bar{d} \times \bar{L}$$

- $\bar{d}$: Average demand per unit time
- $\bar{L}$: Average lead time (same units)
- μ_{DLT} : Expected lead-time demand

Safety Stock Calculation

Safety stock protects against variance in demand and lead time.

$$\text{Safety Stock} = z \cdot \sigma_{DLT}$$

- z : Safety factor (from standard normal table)
- σ_{DLT} : Standard deviation of lead-time demand
- Example: $z = 1.65$ provides a **95%** cycle service level.

Safety Stock & Lead-Time Demand Variance

Calculating uncertainty when both demand and lead time are stochastic variables.



The Combined Variance Formula

$$\sigma_{DLT} = \sqrt{(L^- \cdot \sigma_d^2 + (d^- \cdot \sigma_L)^2)}$$

- σ_d^2 : Variance of demand per unit time
- σ_L^2 : Variance of lead time

Special Cases & Final Reorder Point

Constant Lead Time ($\sigma_L=0$):

$$\sigma_{DLT} = \sigma_d \sqrt{L^-}$$

Constant Demand ($\sigma_d=0$):

$$\sigma_{DLT} = d^- \cdot \sigma_L$$

Full Reorder Point (r):

$$r = d^- L^- + z \sqrt{(L^- \cdot \sigma_d^2 + (d^- \cdot \sigma_L)^2)}$$

Case Study: Solving the Dealership Challenge

Applying combined variance formulas to Julia's Mercedes inventory parameters.



Step 1: Identify Input Parameters

- Demand: $\bar{d} = 9$ cars/mo; $\sigma_d = 1.73$ (from 6-12 range)
- Lead Time: $\bar{L} = 2.5$ mos; $\sigma_L = 0.87$ (from 1-4 range)
- Service Level: 95% $\rightarrow z = 1.65$

Step 2: Calculate Combined Variance (σ_{DLT})

$$\sigma_{DLT} = \sqrt{(\bar{L} \cdot \sigma_d^2 + (\bar{d} \cdot \sigma_L)^2)}$$
$$\sigma_{DLT} = \sqrt{(2.5 \cdot 3 + (9 \cdot 0.87)^2)} \approx 8.3 \text{ cars}$$

Step 3: Determine Reorder Point (r)

$$r = \bar{d} \bar{L} + z \cdot \sigma_{DLT}$$
$$r = (9 \cdot 2.5) + (1.65 \cdot 8.3) \approx 36 \text{ cars}$$

Safety Stock = 13.7 cars

Case Study: Solving the Order Quantity

Determining the optimal number of Mercedes to order for Julia's dealership.



Economic Order Quantity (Q) Recalculation

$Q \approx 4.1$ cars

Using the EOQ formula $Q = \sqrt{(2DS/H)}$ with Julia's parameters ($D=108/\text{yr}$, $S=\$570$, $H=\$7,200/\text{yr}$), the optimal quantity is ~4 cars, not 14.

Why 14 Cars is Inefficient

- **Excessive Holding Cost:** Ordering 14 cars creates a massive average inventory of 7 luxury cars, incurring ~\$4,200/mo in holding costs.
- **Imbalance:** The current policy over-prioritizes reducing order frequency (\$570 setup) at the expense of capital tied up in inventory (\$600/car/mo).
- **Capital Drag:** A lot size of 4 better aligns the inventory flow with the actual monthly demand of 9 cars.

Conclusion: The current 14-car policy is sub-optimal and significantly increases total cost.